

Print at 100% / Actual size, SINGLE-sided. Crease along the grey line, fold so both printed faces are outward, glue, then cut. This bar must measure exactly 80 mm.

[illegible]

BIBIGUBES – BIBIGUBES AND THE MAP

BIG CUBES

4x4 and 5x5

Reduce: build the 6 centers → pair the 12 edges → solve it as a 3x3. Rw = 2 layers wide. 3Rw = 3 layers. 2R = 2nd Layer ONLY – never widen it.

last 2 edges

4x4 PLL parity

4x4 OLL parity – 1 edge pair flipped – half the time

5x5 edge parity – 1 edge pair flipped – half the time – one more differs, 5x5 has no PLL parity

Fit parity during the last layer before OLL and PLL – both parity algorithms move corners

NOTATION



Each letter tells a face clockwise, seen from outside the cube – so you turn your seat left, D and B look counter-clockwise – counter-clockwise = 2 half turns.

M = middle slice, turning the way L, R turn. x y z = the whole cube, turning the way R, U, F turn. Lowercase i = that face plus the layer behind it, turning together.

$$\begin{matrix} R'w & U' & R' & U & F & R & F & R'w \\ 2R2 & U2 & 2R2 & Uw2 & 2R2 & Uw2 \end{matrix}$$
 slice out, run it, slice back – both cubes
 2 edge pairs swapped – half the time

$$\begin{matrix} Rw & U2 & Rw' & U2 & Rw' & U2 & Rw' \\ Rw & U & x & Rw' & U2 & Rw' & U2 & Rw' \end{matrix}$$
 slice out, run it, slice back – both cubes
 2 edge pairs swapped – half the time

BEGINNER SHORTCUT – use it until you know Sune
 Twist a yellow corner four turns at a time, turning ONLY the top face between corners.
 yellow faces RIGHT R'D'R'D x2
 yellow faces FRONT R'D'R'D x2

WORDS

algorithm = a fixed sequence of turns - trigger - a short algorithm your hands run as one unit - sexy move = the R U R' U' trigger - the R U R' U' trigger - parity = a case a 3x3 cannot have: edge pair = two edges glued into one, on a 4x4 or 5x5

RURU' sexy move
 RURU Sune
 R'F'F' sledgehammer
 gap = change grip

[illegible]

ANNEX – BIG CUBES AND THE MAP

BIG CUBES

4x4 and 5x5

Reduce, build 6 centers – pair the 12 edges –> solve it as 3x3. Rw = 2 layers wide. 3Rw = 3 layers. 2R = 2nd Layer ONLY – never widen it.

last 2 edges

4x4 PLL parity

$$\begin{matrix} R'w & U & R' & U & R' & F & R & F \\ 2R2 & U2 & 2R2 & Uw2 & 2R2 & Uw2 \end{matrix}$$

slice out, run it, slice back – both cubes

2 edge pairs swapped - half the time

4x4 OLL parity – 1 edge pair flipped - half the time

$$\begin{matrix} R & W & U2 & x & R & W & U2 & R'w \\ 555 \text{ edge parity} \end{matrix}$$

edge pair flipped - half the time - one U2 differs, 5x5 has no PLL parity

5x5 edge parity

$$\begin{matrix} R'w & U & x & R & W & U2 & R'w \\ 555 \text{ edge parity} \end{matrix}$$

U2 Lw Lw U2 R'w

slice out, run the last layer before OLL and PLL – both parity algorithms move corners

NOTATION



Each letter turns a face clockwise, seen from outside the cube – so then you see let's L and B look counter-clockwise –> counter-clockwise, = 2 half turn.

M – middle slice, turning the way L turns.

x, y, z = the whole cube, turning the way R, U, F turn. Lowercase if it has that face plus the layer behind it, turning together.



Twist a yellow corner four turns at a time, turning ONLY the top face between corners.

yellow faces RIGHT

yellow faces FRONT

$$\begin{matrix} D'R'DR & x2 \\ D'R'DR & x2 \end{matrix}$$

BEGINNER SHORTCUT – use it until you know Sune

Sune = twist a yellow corner four turns at a time, turning ONLY the top face between corners.

[GOO](#)
[A](#)

WORDS

algorithm = a fixed sequence of turns - trigger = a short turn your hands run as one unit - sexy move = the R U R' U' trigger - the R' F R' U' trigger, 3x3 as a case 3x3 cannot have edge pair = two edges glued into one, on a 4x4 or 5x5

RURU sexy move **RURU** sexy **R'F'F'** sldedgehammer gap = change grip

[illegible][illegible]

ANNEX – BIG CUBES AND THE MAP

BIG CUBES 4x4 and 5x5
 Reduce; build the centers → pair the 12 edges → solve it as a 3x3. Rw = 2 layers wide. 3Rw = 3 layers. 2R = 2nd Layer ONLY → never widen it.

last 2 edges
 4x4 PLL parity
 $R'w \ U' \ R' \ U \ R' \ F \ R' \ F \ R'$ slice out, run it, slice back – both cubes
 $2R^2 \ U^2 \ 2R^2 \ U^2 \ 2R^2 \ U^2$ 2 edge pairs swapped - half the time

4x4 OLL parity – 1 edge pair flipped - half the time
 5x5 edge parity – 1 edge pair flipped - half the time - one move differs; 5x5 has no PLL parity

Rw U2 x Rw U2 Rw U2 Rw U2 Rw U2 Rw U2 Rw U2 Rw U2
 5x5 last layer – all faces are solved, but both sides algorithms may turn corners

NOTATION
 Each letter turns that face clockwise,
 seen from outside the cube – so from your seat L, D and B look counter-clockwise – counter-clockwise = 2 half turn.
 M = middle slice, turning the way L turns. x y z = the whole cube, turning the way R, U and F turn. Lowercase i f = that face plus the layer behind it, turning together.

WORDS
 algorithm → a fixed sequence of turns - trigger = a short algorithm you hands run as one unit - sexy move – the R U R' U' trigger – sledgehammer – the R' F R' trigger. parity is a case a 3x5 cannot have - edge pair = two edges glued into one, on a 4x4 or 5x5

RURU sexy move **RURU**Sexy **R'FRF'** sledgehammer case - glue chips

ANNEX – BIG CUBES AND THE MAP DOD A 353

BIG CUBES 4x4 and 5x5

Reduce, build the 6 centers – pair the 12 edges – solve it as a 3x3. R.w = 2 layers wide. 3.Rw = 3 layers. 2.R = 2nd Layer ONLY – never build the 3rd layer.

last 2 edges

4x4 PLL parity

$$\begin{matrix} R'w & U' & R' & U & R' & F & R' & F & R' & U' \\ 2R'2 & U'2 & 2R'2 & U'2w & 2R'2 & U'2w & & & & \end{matrix}$$

2 edge out, run it, slice back – both cubes
slide pair, swapped – half the time

4x4 OLL parity – 1 edge pair flipped – half the time

$$\begin{matrix} R'w & U'2 & x & R'w & U'2 & R'w & U'2 & R'w' & U'2 & L'w & R'w & U'2 & R'w' & U'2 & R'w' \\ 5x5 \text{ edge parity} & 1 \text{ edge pair flipped – half the time – one move differs, 5x5 has no PLL parity} \end{matrix}$$

5x5 edge parity – 1 edge pair flipped – half the time

$$\begin{matrix} R'w & U'2 & x & R'w & U'2 & R'w & U'2 & 3R'w & U'2 & L'w & U'2 & R'w & U'2 & R'w' & U'2 & R'w' \\ \text{5x5 center parity} & \text{the last 3 layer, 5x5 has no PLL parity} \end{matrix}$$

NOTATION

Each letter turns that face clockwise, seen from outside the cube – so from your seat L, D and B look counter-clockwise – counter-clockwise = 2 half turn.

M = middle slice, turning the way L turns. x y z = whole cube, turning the way R U L and F turn. Lowercase i f = that face plus the layer behind it, turning together.

WORDS

wordform = a fixed sequence of turns - trigger = a short algorithm that hands runs as one unit - sexy move = the R U R' U' trigger - wordform + the R' F R' F' trigger. parity = a case a 3x5 cannot have - edge pair = 2 edges glued into one, on a 4x4 or 5x5

RURU sexy move **RURU** Sexu **R'FRF** sledgehammer gap = change chirality